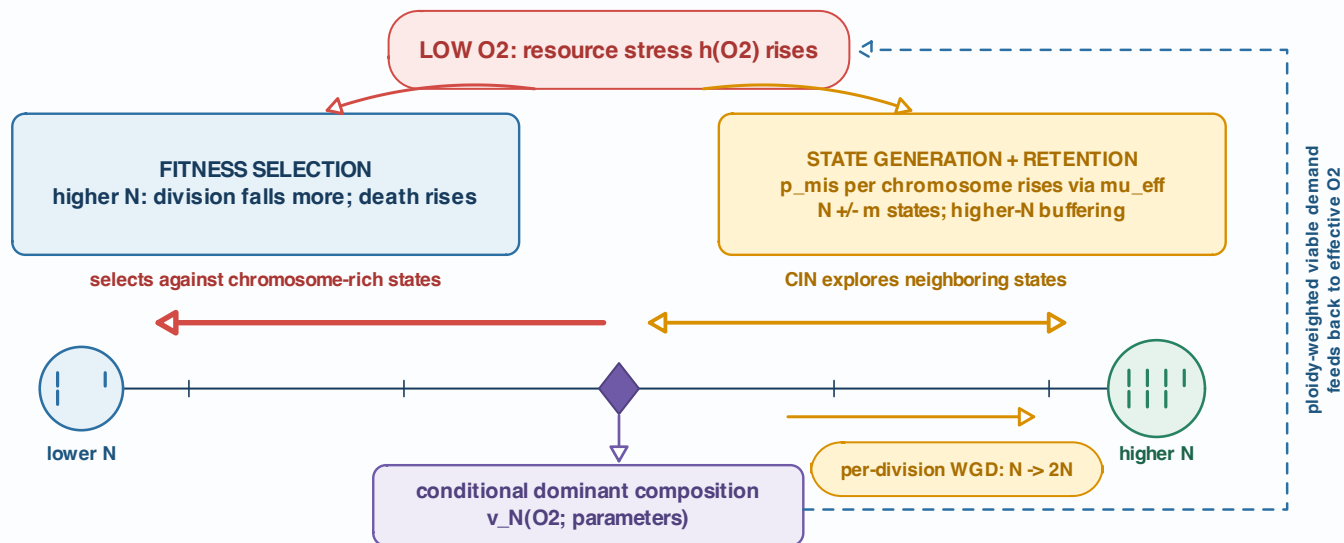


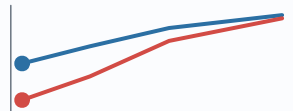
A Resource limitation couples opposing pressures on ploidy evolution



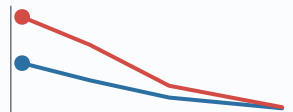
B Low- O_2 fitness selection

— lower N — higher N

Effective division rate



Hypoxia-associated death

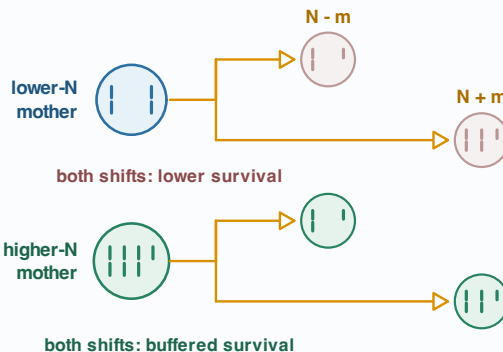


O2 increasing

Low O_2 favors lower-N expansion

C State generation + survival filtering

p_{mis} per chromosome rises
via μ_{eff}



WGD: $N \rightarrow 2N$
constant per division

out-of-grid offspring
enter nonviable pool

D Population integration

fitness + CIN/WGD + survival

fixed- O_2 operator
 $M(O_2)$

conditional dominant composition v_N



chromosome state N

mean ploidy = $\sum(N v_N) / 22$

interpret when the spectral gap resolves v_N